

All figures below are computed on **synthetic data** — the assignment CSVs plus a hand-built DHIS2 sample — not real Rwanda health records. Facility mortality of 14.3–73.2 per 1,000 live births runs three to four times real Rwandan rates.

The choice: Problem A, and why not B or C

OPTION	VERDICT	WHY
A — Quarterly Health Bulletin, automated	Chosen	Only option finishable and provable in 6 weeks; operates on data that already arrives, in a shape that already exists. The trust wedge for later work.
B — Real-time facility dashboard	Out of reach <i>this sprint</i>	Needs facility-level freshness. Reporting substrate is monthly aggregates on a 2–3wk lag — a granularity/cadence gap, not a coverage gap, regardless of how many facilities report.
C — Unified patient view (TB/HIV)	Rejected	No shared patient ID between the TB and HIV systems, so records are matched by probability. A wrong match puts one patient's HIV status into another's treatment decision, and no MoH process exists to review and sign that risk off inside 6 weeks. Propose a weekly reconciliation report instead: flags likely co-infection for human review, no record merging, no clinical claim.

What was built

A pipeline that reads the five provided CSVs plus a DHIS2-shaped sample, resolves every field through a 71-row declared crosswalk, and publishes all four 2024 quarters as self-contained HTML with **zero client-side JavaScript**. Python + Hamilton + DuckDB + Parquet (bronze → silver → gold → checks); Astro + Vega-Lite compiled to static SVG for render. **5 automated checks** gate publication, each one added because it caught a real defect. Live: sand-fde-bulletin.pages.dev · Repo: github.com/nyashkn/sand-fde-int

Two findings that shaped it more than any metric in the brief

- 1 — **The largest cause of neonatal death isn't a clinical cause.** death_other is **30.7%** of resolved deaths, the largest single bucket in all ten held months (28.8%–32.1%). The two named causes behind it can't be separated: birth asphyxia **23.7%** vs prematurity **23.6%**, a gap of 15 deaths in 9,036 — and which one leads flips between months. A bulletin ranking named clinical causes would be reporting noise. The finding is about the reporting system, not newborn medicine.
- 2 — **A striking correlation is a confound.** A composite equipment index correlates ≈ -0.87 with facility mortality, pooled across 117 facilities — reads as "equip facilities, save newborns." Stratified by tier it collapses to near zero: **District -0.07 , Health Centre $+0.04$, Provincial -0.29** . The pooled number describes which tier a facility sits in, not what equipping it would do. Publishing the pooled figure alone would have pointed a Ministry's budget at a confound; the bulletin shows every covariate pooled *and* within tier, caveat attached, not withheld.

"Our data is a mess" is a symptom report, not a problem statement

FAILURE CLASS	MEANING	IMPLIES
Absence	Never captured at all	Digitisation problem
Latency	Captured, arrives too late to act on	Pipeline problem — DHIS2 runs 2–3wk behind
Distrust	Arrives on time, nobody believes it	Provenance problem — sources disagree, nobody reconciles
Last mile	Believed, never reaches a decision	Workflow problem

Week 1's job is to find which dominates. Distrust is tested *first* — cheapest to check, most commonly missed — not because the brief's sentence says so. One question for every interview, because the answers won't match: "*When two systems give different numbers, what happens?*"

Who I'd talk to, in what order (Week 1)

DAY	WHO	THE QUESTION THAT MATTERS
1	Solutions Manager, Country Director	Why were the use cases chosen <i>before</i> discovery?
1–2	MoH Director (sponsor)	"Tell me about a decision last quarter you'd have made differently with better data." Then: " <i>When the numbers you're shown disagree with what you believe, which do you act on?</i> "
2	The bulletin analyst	Not "what do you need" — " <i>show me last quarter's, and open the files you built it from.</i> "
2–3	DHIS2/HMIS focal point	Where does the 2–3wk delay actually accrue, and how do paper facilities report today?
3	2 district health officers (1 strong, 1 weak district)	" <i>What did you look at immediately before your last resource decision?</i> "
3–4	1 hospital, 1 clinic, 1 paper-only facility — the data clerk , not the medical director	Watch month-end reporting happen
4	MoH ICT / InfoSec	Hosting, data residency, approval lead times — the classic 6-week killer

Every question is **counterfactual** ("what did you do last month") or **observational** ("show me"), never **solicitational** ("what would you like") — solicitation just returns a feature list.

Top assumptions and risks (of 9 tracked in full)

#	ASSUMPTION	IF FALSE
A3	Reporting labour is mostly mechanical	Automate assembly only; reset the ROI target publicly
A7	A named DHO with cleared hours exists	Downgrade honestly; stop calling it capacity transfer
A8	The bulletin is read by someone who decides	Stop. Find what is read
A10	Hosting / InfoSec approval achievable in ≤3 weeks	Re-plan around Ministry-owned infrastructure

The Week 1 gate. The commitment to A is conditional on five gates assessed at end of Week 1, not final at proposal time: **G1** Value (a decision-maker used the last bulletin for something identifiable) · **G2** Mechanisability (≥60% of cycle time is mechanical) · **G3** Inputs (DHIS2 access confirmed with a stated cutoff) · **G4** Runtime (a place to run and hand over exists) · **G5** Operator (a named receiver is assigned). G1 or G3 failing means A is the wrong commitment — said in Week 1, not Week 5.

What would change my mind

OBSERVATION	RESPONSE
Districts decide weekly and can't today	Re-scope toward B for adequate-granularity facilities, honestly named partial
The bulletin is not read by any decision-maker	Stop. Find what is read
Co-infected patients harmed today , measurably	Escalate C as clinical safety, resourced properly
Hosting/InfoSec approval takes >3 weeks	Re-plan around an offline deliverable; escalate Day 3

How this survived review. The problem selection was prompted hostilely against three independent model families (GPT-5.6-sol, Gemini 3.1 Pro, Grok 4.5). All three independently chose **Problem A** — but rejected the original justification, finding errors unanimously (an impossible 5-day publication target given a 2–3wk upstream lag; an overstated ROI baseline; a false claim that A's mart is "most of" B's data layer). All corrected in the full document.